Kinematics and Trajectory Planning of a PRRR Manipulator for Car Surveillance Robot

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***Abstract*—This work presents the design, kinematic modeling, and trajectory planning of a PRRR manipulator integrated into an autonomous car-surveillance robot intended for under-vehicle inspection and maintenance. Existing systems are too limited: restricted mobility, operator dependence and poor adaptability, this PRRR system enables compact navigation and manipulation beneath a standard vehicle. A modified Denavit–Hartenberg formulation is used to derive forward kinematics, while inverse kinematics equations yield up to four feasible joint solutions for arbitrary target positions. Workspace characterization gives the range of motion and workspace attained by the robot. Minimum-jerk trajectory simulations demonstrate smooth and dynamically feasible motion between boundary configurations, validating the robot's capability for continuous inspection tasks. The results support the feasibility of PRRR-type manipulators for autonomous car-surveillance applications requiring precision, reachability, and smooth actuation.**

I. Introduction

The rapid development of robotics offers compelling solutions for automated inspection and maintenance, particularly in challenging environments like a vehicle's undercarriage system. This paper details the design, implementation, and kinematic analysis of an autonomous car surveillance robot system developed for diagnostic and maintenance purposes. This robot is specifically engineered to improve the efficiency and convenience of undercarriage inspection by eliminating the need for extensive human intervention.

Current vehicle inspection methods often rely on heavy equipment and human operators, leading to accessibility issues and similar limitations [1]. The motivation for this project is from existing research on Vehicle Undercarriage Inspection Robots (TP X LTA). The system had key limitations like poor adaptation to different terrain and the device required continuous human assistance to maneuver the device for complete inspection.

To address these shortcomings, the objective of this project was to build a PRRR manipulator with an end-effector for comprehensive car surveillance and to facilitate minor repair and maintenance tasks. The robot must be compact, robust, and capable of autonomous navigation and fine-grained manipulation within the confined space beneath a standard vehicle.

II. Design

A. Forward Kinematics

To describe the motion of the camera mounted at the tip of the mechanism, we model the structure as a 4-DOF PRRR chain consisting of [2]

Fig. 1. Photo-rendered image of the car surveillance robot system created using SOLIDWORKS Visualize

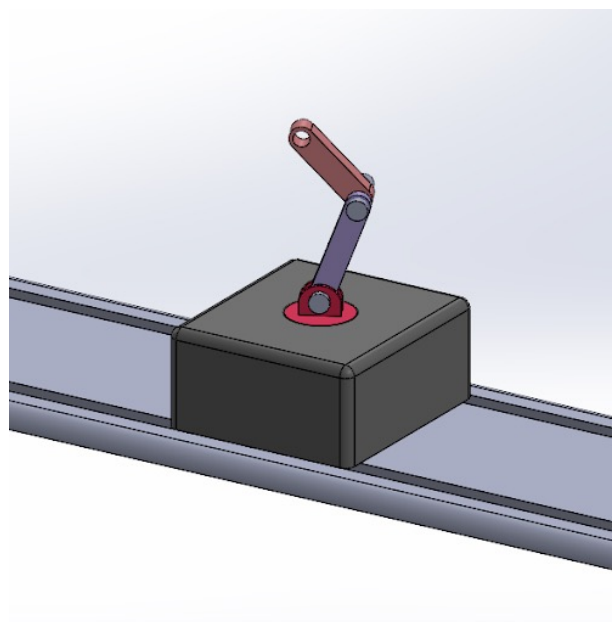

Fig. 2. Image of the SOLIDWORKS Model of the PRRR Manipulator with an empty link for detachable end-effector

- Joint 1: prismatic motion sss along the horizontal rail
- Joint 2: revolute motion q_1 about the global Z-axis
- Joint 3: revolute motion q_2 about a rotated Y-axis.
- Joint 4: revolute motion q_3 about a rotated X-axis

The generalized coordinate vector is

$$\mathbf{q} = \begin{bmatrix} s & q_1 & q_2 & q_3 \end{bmatrix}^T$$

We adopt the modified Denavit-Hartenberg (MDH) convention. In terms of elementary transformations, the homogeneous transform from the base frame to the camera frame can be written as

$$T(\mathbf{q}) = T_y(s)R_z(q_1)T_z(h)R_y(q_2)R_x(\alpha)T_z(L_1)R_y(\beta)T_z(L_2)$$

$$T(\mathbf{q}) = \begin{bmatrix} \mathbf{R}(\mathbf{q}) & \mathbf{p}(\mathbf{q}) \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \quad \mathbf{p}(\mathbf{q}) = \begin{bmatrix} p_x(q_1, q_2, q_3) \\ s + p_y(q_1, q_2, q_3) \\ p_z(q_1, q_2, q_3) \end{bmatrix}$$

For compactness we introduce the shorthands:

$$\begin{aligned} S_1 &= \sin q_1, & C_1 &= \cos q_1 \\ S_2 &= \sin q_2, & C_2 &= \cos q_2 \\ \gamma &= \alpha + \beta + q_3, & S_\gamma &= \sin\gamma, \quad C_\gamma = \cos\gamma \\ A &= L_1 \sin\alpha + L_2 \sin\gamma, \\ B &= L_1 \cos\alpha + L_2 \cos\gamma \end{aligned}$$

$$J_v(\mathbf{q}) = \begin{bmatrix} 0 & -(BC_1 + AS_1S_2) & AC_1C_2 & L_2(S_1S_\gamma + S_2C_1C_\gamma) \\ 1 & -(BS_1 - AC_1S_2) & AS_2C_2 & L_2(S_1S_\gamma - S_2C_1C_\gamma) \\ 0 & 0 & -AS_2 & L_2C_2C_\gamma \end{bmatrix}$$

B. Inverse Kinematics

The inverse-kinematics (IK) solution computes the three joint variables q_1 (prismatic along $+Y$), q_2 (yaw about $+Z$), and q_3 (pitch about $+X$) that position the end-effector at a desired point $P = (x,\, y,\, z)$ in a global Cartesian frame. When the shoulder joint is tilted by an angle α, the vertical geometry of the arm satisfies

$$z = H + (L_2 + L_3 \cos q_3)\cos\alpha,$$

where $H =$ rail height + block height and L_2, L_3 are the link lengths. Solving for q_3 yields

$$\cos q_3 = \frac{(z - H)/\cos\alpha - L_2}{L_3}, \qquad q_3 = \pm\arccos(c_3).$$

For each q_3, the horizontal projection of the mechanism in the tilted frame becomes

$$x_0 = \sin\alpha\,(L_2 + L_3\cos q_3), \qquad y_0 = -L_3 \sin q_3,$$

with horizontal reach

$$r = \sqrt{x_0^2 + y_0^2}.$$

The yaw joint q_2 rotates this projected vector so that its x-projection matches the target x, imposing the constraint

$$x = r\cos(\delta + q_2), \qquad \delta = \operatorname{atan2}(y_0,\, x_0),$$

yielding two yaw solutions

$$q_2 = -\delta \pm \arccos\left(\frac{x}{r}\right).$$

Finally, the prismatic joint follows from the y-direction balance:

$$q_1 = y - (s_2 x_0 + c_2 y_0),$$

Taken together, these expressions produce up to four IK solutions corresponding to combinations of elbow up/down and sweeping left/right yaw configurations.

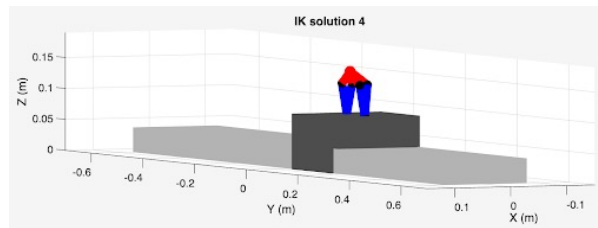

Fig. 3. Four inverse-kinematics solutions for reaching the end-effector position $P = [0.01,\ 0.1,\ 0.16]^T$.

C. Singularity Analysis

The singular configurations of the PRRR manipulator are determined by examining the conditions under which the Jacobian matrix $J(q)$, which maps joint velocities $\dot{\mathbf{q}}$ to Cartesian end-effector velocities $\dot{\mathbf{p}}$, loses rank. Since the prismatic joint q_1 produces a constant translational velocity along the global Y-axis, singular behavior arises primarily from the coupling between the rotary joints q_2 and q_3. Differentiation of the forward-kinematics expression for the end-effector position yields a Jacobian whose determinant can be written as

$$\det J(q) \propto \sin q_3 \left(-A \sin q_2 + B \cos q_2 \right),$$

where

$$A = \sin\alpha\,(L_2 + L_3\cos q_3), \qquad B = L_3 \sin q_3,$$

and α denotes the fixed shoulder-tilt angle. We suspected two principal classes of singularities follow from the condition $\det J(q) = 0$.

When $\sin q_3 = 0$, i.e., $q_3 = k\pi$, the elbow becomes fully extended or fully folded in the tilt plane. In this configuration, variations in q_3 no longer generate independent Cartesian motion, reducing the manipulator's instantaneous mobility.

A second singular set occurs when

$$\tan q_2 = \frac{B}{A} = \frac{L_3 \sin q_3}{\sin\alpha\,(L_2 + L_3\cos q_3)}.$$

In these configurations, the velocity contributions of the shoulder and elbow joints become linearly dependent, causing the Jacobian to drop rank and eliminating one direction of achievable Cartesian motion.

These singularities represent configurations in which commanded end-effector velocities may require arbitrarily large joint velocities or become unachievable altogether. Identifying and avoiding these sets is therefore essential for maintaining manipulability, ensuring numerically stable control, and guaranteeing safe trajectory execution.

III. Simulation Results

A. Workspace Characterization

The robot workspace is generated by sweeping the prismatic joint $q_1 \in [-0.6, 0.6]\,\mathrm{m}$, together with the revolute joints $q_2 \in [-180°, 180°]$ and $q_3 \in [-135°, 135°]$ over their full ranges without self-collision. The resulting set of reachable end-effector points forms a dense three-dimensional cloud. The structure of the cloud reveals an elongated,tube-like reachable region dominated by the translational motion of the prismatic axis, while the rotational joints primarily determine the radial distribution and thickness of the workspace volume.

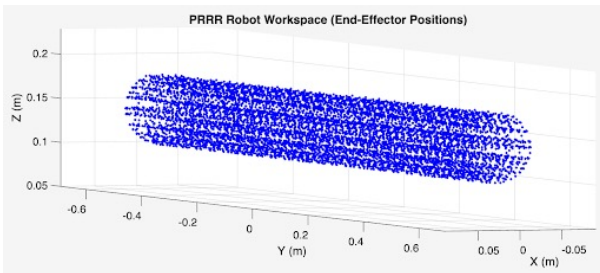

Fig. 4. Reachable workspace of the PRRR robot characterized by exhaustively sweeping all joint degrees of freedom and computing the corresponding end-effector positions.

B. Trajectory Analysis

To evaluate the dynamic capabilities of the PRRR manipulator, a minimum-jerk trajectory was generated to move between two extreme joint configuration

$$q_{\text{start}} = [-0.6,\ -\pi,\ -3\pi/4], q_{\text{end}} = [0.6,\ \pi,\ 3\pi/4]$$

over a duration of $1.0\,\mathrm{s}$ with discretization of time $\Delta t = 0.01\,\mathrm{s}$. The resulting Cartesian motion (Fig 5) demonstrates a smooth transition of the end-effector between the two boundary poses while respecting the tilt-adjusted kinematic structure of the robot. Corresponding joint trajectories (Fig 6) exhibit the expected minimum-jerk structure, with position profiles that are continuous and monotonic, velocity profiles that rise and fall symmetrically, and accelerations that start and end at zero, ensuring dynamic feasibility and actuator-friendly motion. This analysis confirms that the current PRRR mechanism can execute large-range motions across its full joint space while maintaining smooth kinematic behavior suitable for real-time control applications.

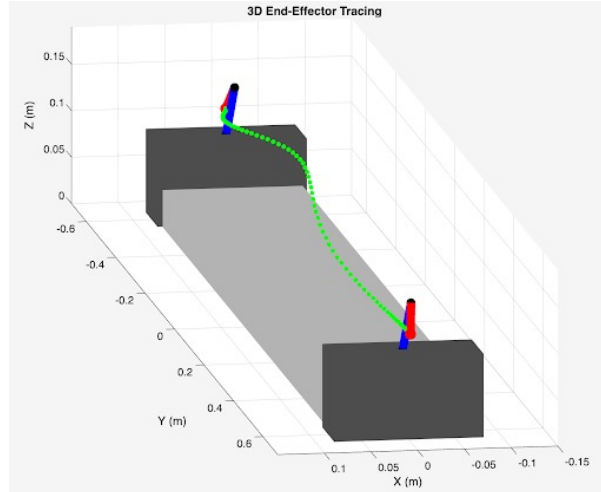

Fig. 5. End-effector trajectory for a minimum-jerk motion from the initial joint configuration $[-0.6,\ -\pi,\ -3\pi/4]^T$ to the final configuration $[0.6,\ \pi,\ 3\pi/4]^T$.

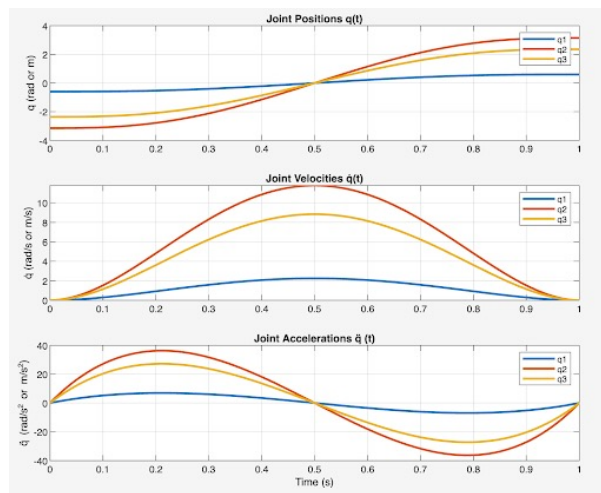

Fig. 6. Joint-space kinematic profiles (position, velocity, acceleration) for the minimum-jerk motion between $[-0.6,\ -\pi,\ -3\pi/4]^T$ and $[0.6,\ \pi,\ 3\pi/4]^T$.

IV. Conclusion

In this study, a Kinematics and Trajectory Planning of a PRRR Manipulator for Car Surveillance Robot was developed and simulated. The system addresses major shortcomings of existing inspection platforms by enabling compact under-vehicle access, full-space positioning, and autonomous maneuvering without constant human assistance. Forward and inverse kinematic analysis confirmed that the mechanism can reach the whole workspace beneath a vehicle, providing optimal coverage for both cameras and tools to ensure effective inspection. The analysis shows that the system can execute its mission in the whole workspace while maintaining a smooth kinematic behavior suitable for real time applications. Moreover, MATLAB simulations confirm the robot's trajectory planning is accurate and feasible for real-world deployment, which is necessary to deploy this system. Overall, the results strengthen the case for PRRR-based architectures in automated inspection and suggest that future work should incorporate dynamic modeling, physical prototyping, and integrated sensing to advance toward operational field use, with the least amount of human intervention. Integrating a tool holding gripper for minor fixes as the end-effector and compactifying the device to accommodate cars with lower ground clearance could be great additions to make this system even more interesting and scalable.

REFERENCES

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[2] M. W. Spong and M. Vidyasagar, *Robot Dynamics and Control*. Wiley, 1989. ISBN: 978-0471612438.